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BS. Zoology 7th

Biostatistics

Relation b/w Varience and Standard Deviation:

- Varience is the square root of standard deviation.
- Standard deviation is the positive square root of the variance.

Example: If,

i) $S.D = 4$.

$\Rightarrow$ Variance $= (4)^2 = 16$

If,

Variance $= 25$

$S.D = \sqrt{25} = 5$.

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Cofficient of Variance:

formula = $C.V = (S.D / Mean \times 100)\%$

C.V is the relative measure of dispersion / Variability.

It is the pure number (unitless) because both S.D and mean have the same units, which cancel out.

Uses: Following uses are:

1. To measure or compare variability amonog data sets with different units of measurements.

2. To check consistency in data:
   Lower C.V = More consistency
   Higher C.V = More Consistency.

3. C.V describes S.D as a % with respect to its mean.

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Standard Deviation: Is defined as:

It is an absolute measure of dispersion. It has the same units as of data.

Standard error of a mean:

$$S.E.M = \sigma / \sqrt{n}.$$

Standard error is the S.D of a sample statistics (like the sample mean).

Estimator of this error is called standard error.

How to reduce S.E.M?

→ By increasing the sample size since S.E.M is inversely proportional to $\sqrt{n}$.

Past Paper's

In following groups of data which is less variable and why?

10, 60, 50, 40, 30, 20	35, 45, 30, 35, 40, 25

Step #:01 Find out mean of each:

$$\bar{X}_A = \frac{10+60+50+40+30+20}{6} = \frac{210}{6} = 35$$

$$\bar{X}_B = \frac{35+45+30+35+40+25}{6} = \frac{210}{6} = 35$$

Step#:02 Find Variance:

For A:

$$(10-35)^2 + (60+35)^2 + (50-35)^2 + (30-35)^2 + (\overset{40-35}{\cancel{30-40}})^2 + (20-35)^2 =$$

$$625 + 625 + 225 + 25 + 25 + 225 = 1750$$

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$$A^2 = \frac{1750}{6} = 291.67.$$

~~Step #:03 Compare Variability~~

For B

$$(35-35)^2+(40-35)^2+(30-35)^2+(35-35)^2$$
$$+(40-35)^2+(25-35)^2$$

$$= 0+100+25+0+25+100 = 250$$

$$= \frac{250}{6} = 41.67.$$

Step #:03 Compare Variability

Variance of A (291.67) is much higher than that of B (41.67).

Hence, B is less variable because it has smaller variance (and S.D).

~(Q # 02)~

Number of

The number of stories in the 13 tallest building for 2 different cities is listed below:

Data Given:

Houston: 75, 71, 64, 56, 53, 55, 47, 55, 52, 50, 50, 50, 47

Pittsburgh: 64, 54, 40, 32, 46, 44, 42, 41, 40, 40, 34, 32, 30

Find mean; $(\bar{X}_H)$:

$$\frac{75 + 71 + 64 + 56 + 53 + 55 + 47 + 55 + 52 + 50 + 50 + 50 + 47}{13} = \frac{725}{13} = 55.76$$

$$(\bar{X}_P) = \frac{64 + 54 + 40 + 32 + 46 + 44 + 42 + 41 + 40 + 40 + 34 + 32 + 30}{13} = \frac{539}{13} = 41.46$$

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Step #:2 Variance $\left(= \frac{\sum_i (X - \bar{X})^2}{n} \right)$

$$X_H^2 = (75 \cdot 55.77)^2 + (71 - 55.77)^2 + (64 - 55.77)^2 + (56 - 55.77)^2 + (53 - 55.77)^2 + (55 - 55.77)^2 + (47 - 55.77)^2 + (55 - 55.7)^2 + (52 - 55.7)^2 + (50 - 55.77)^2 + (50 - 55.7)^2 + (50 - 55.7)^2 + (47 - 55.77)^2 \,/\, 13$$

$$= \frac{1043.92}{13} = 80.30.$$

$$XP^2 \, (64 - 41.46)^2 + (54 - 41.46)^2 + (40 - 41.46)^2 + (32 - 41.46)^2 + (44 - 41.46)^2 + (42 - 41.46)^2 + (41 - 41.46)^2 + (40 - 41.46)^2 + (40 - 41.46)^2 + (34 - 41.46)^2 + (34 - 41.46)^2 + (34 - 41.46)^2 + (32 - 41.46)^2 + (30 - 41.46)^2 \,/\, 13$$

$$- \frac{1523.84}{13} - 117.22.$$

Pittsburgh's data is more variable because it has a larger variance and S.D than houston.